

A skew INAR(1) process on $\mathbb{Z}$

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Abstract Integer-valued time series models have been a recurrent theme considered in many papers in the last three decades, but only a few of them have dealt with models on $\mathbb{Z}$ (that is, including both negative and positive integers). Our aim in this paper is to introduce a first-order, integer-valued autoregressive process on $\mathbb{Z}$ with skew discrete Laplace marginals (Kozubowski and Inusah, Ann Inst Stat Math 58:555–571, 2006). For this, we define a new operator that acts on two independent latent processes, similarly as made by Freeland (Adv Stat Anal 94:217–229, 2010). We derive some joint and conditional basic properties of the proposed process such as characteristic function, moments, higher-order moments and jumps. Estimators for the parameters of our model are proposed and their asymptotic normality is established. We run a Monte Carlo simulation to evaluate the finite-sample performance of these estimators. In order to illustrate the potential for practice of our process we apply it to a real data set about stock market.

Keywords Integer-valued time series models · Skew discrete Laplace distribution · Latent process · Thinning operator · Estimation · Asymptotic normality

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1 Introduction

Count time series models have been a recurrent theme considered in many papers in the last three decades. Pioneering works in this interesting theme are due to [Steutel and van Harn \(1979\)](#), [McKenzie \(1985\)](#), [Al-Osh and Alzaid \(1987\)](#) and [McKenzie \(1988\)](#). From these works, count valued time series models with Poisson marginals may be obtained. Issues such as inference and forecasting for Poisson time series models have been discussed by [Freeland and McCabe \(2004a\)](#) and [Freeland and McCabe \(2004b\)](#). Asymptotic properties of estimators for a Poisson AR(1) model were established by [Freeland and McCabe \(2005\)](#).

There are several variants of Poisson INAR(1) models; for instance, see the surveys by [Weiß \(2008\)](#) and [Fokianos \(2011\)](#). One of these variants was introduced and studied by [Ristić et al. \(2009\)](#). They constructed and studied several properties of a stationary INAR(1) process with geometric marginals based on a negative binomial thinning operator; this model is named new geometric INAR(1) process (in short NGINAR). Further results on this model can be found for instance in [Bakouch \(2010\)](#). The NGINAR(1) model is overdispersed and therefore it is an alternative to the Poisson AR(1) model when one is analyzing time series data that present overdispersion.

On the other hand, only a few papers have dealt with time series models on $\mathbb{Z}$ (that is, including both negative and positive integers). Such a models can arise naturally in practical situations. For example, it is frequent to encounter a non-stationary count time series. Particularly, this happens with count series which are small in value and show a trend having relatively large fluctuation. To handle such a non-stationary series, the difference operator is commonly applied in the series to achieve stationarity; for instance, see [Kim and Park \(2008\)](#) and [Zhang et al. \(2010\)](#). The differenced series may contain negative integer values and therefore the usual count models are not able to fit these data. With this in mind, some models were proposed in the literature. [Kim and Park \(2008\)](#) proposed an integer-valued autoregressive process on $\mathbb{Z}$ based on a signed binomial thinning. This process was recently generalized by [Zhang et al. \(2010\)](#).

INAR models on $\mathbb{Z}$ can also occur naturally in finance and economics. One can be interested in modeling the change in price of a commodity, stock or other financial product. The price change is treated commonly with the states: no price change, up one tick, down one tick, up two ticks, down two ticks etc. Then, the price change can be viewed as a stochastic process assuming values on $\mathbb{Z}$ with j meaning up j ticks if $j > 0$ and down j ticks if $j < 0$. For instance, see Application's Section of [Freeland \(2010\)](#).

In a different approach of that considered by [Kim and Park \(2008\)](#), [Freeland \(2010\)](#) introduced a stationary AR(1) process on $\mathbb{Z}$ with symmetric Skellam marginals (which are distributed as a difference between two iid Poisson random variables); this model is named true INAR(1) process (in short TINAR). The idea of Freeland was to define a modified binomial thinning operator, which involves two iid latent Poisson AR(1) processes.

Our aim in this paper is to introduce a stationary INAR(1) process on $\mathbb{Z}$ with skew discrete Laplace marginals ([Kozubowski and Inusah 2006](#)). We named this model by skew INAR(1) process on $\mathbb{Z}$ (in short SINARZ). For this, we propose a modified version of the negative binomial thinning operator in a similar fashion as made by

Freeland (2010). Here, our thinning operator acts on two independent but not necessarily identically distributed latent NGINAR(1) processes. The skew discrete Laplace (SDL) distributions (and other similar distributions) have a great importance in analysis of hydroclimatic episodes such as droughts, floods and El Niño; for instance, see the introduction section of Kozubowski and Inusah (2006). Due to these interesting applications of the SDL distribution, we think that our SINARZ(1) process can also be of great interest in these areas when there is a temporal dependence.

We have some advantages of our model with respect to that one given in Freeland (2010). The main of them are:

- *Accommodation of skewness* (the TINAR(1) process is symmetric);
- *Mathematical simplicity of our model*. For instance, the probability and distribution functions of the skew discrete Laplace distribution have a simple form (see below) in contrast with the Skellam distribution which has associated probability function involving the modified Bessel function of the first kind.

The present paper is organized in the following way. In Sect. 2 we introduce our skew INAR(1) process on $\mathbb{Z}$. We also obtain some statistical properties such as joint moments and joint and conditional characteristic functions among others. In Sect. 3, we obtain joint higher-order moments and present some properties of the jumps for our SINARZ(1) process. Estimators for the parameters of our model and their asymptotic normality are presented in Sect. 4. We also present some simulation results in order to evaluate the finite-sample performance of the proposed estimators. In Sect. 5 we illustrate the potential for practice of our model by applying it to a real data set about stock market.

2 SINARZ(1) process

In this section we introduce a stationary first-order integer-valued autoregressive process on $\mathbb{Z}$ with skew discrete Laplace marginals, named skew INAR(1) model on $\mathbb{Z}$ (in short SINARZ). For this, we first introduce some notation and the NGINAR(1) model by Ristić et al. (2009).

Let $\alpha \in [0, 1)$ and “ $*$ ” be the negative binomial thinning operator (Ristić et al. 2009), which is defined by

$$\alpha * X = \sum_{i=1}^X W_{iX},$$

for $X \in \mathbb{N}$ and $\alpha * 0 = 0$, where $\{W_{ij}\}_{i,j \in \mathbb{N}}$ is a sequence of iid random variables following a geometric distribution on $\mathbb{N}$ with mean α .

Definition 1 (NGINAR(1) process) Let $\{X_t\}_{t \in \mathbb{N}}$ be a stationary process having geometric marginals with probability function assuming the form $P(X_t = x) = \mu^x / (1 + \mu)^{x+1}$, where $x \in \mathbb{N}$ and $\mu > 0$. The NGINAR(1) process is defined by

$$X_t = \alpha * X_{t-1} + \epsilon_t, \quad t \in \mathbb{N}^*,$$

for $\alpha \in [0, \mu/(1 + \mu))$, where $\{\epsilon_t\}_{t \in \mathbb{N}^*}$ is a sequence of iid random variables independent of $\{W_{ij}\}$, and ϵ_t and X_{t-l} are independent for all $l \geq 1$; here $\mathbb{N}^* \equiv \mathbb{N} \setminus \{0\}$.

Ristić et al. (2009) showed that the probability function of ϵ_t is given by

$$P(\epsilon_t = l) = \left(1 - \frac{\alpha\mu}{\mu - \alpha}\right) \frac{\mu^l}{(1 + \mu)^{l+1}} + \frac{\alpha\mu}{\mu - \alpha} \frac{\alpha^l}{(1 + \alpha)^{l+1}}, \quad l \in \mathbb{N}, \quad (1)$$

that is, the random variable ϵ_t is a mixture of two independent random variables that follow geometric distributions with means μ and α .

We now present briefly the skew discrete Laplace (SDL) distribution studied in Kozubowski and Inusah (2006), which will be the marginal of our process. A discrete random variable Z following a SDL distribution with parameters $\mu_1 > 0$ and $\mu_2 > 0$ has probability and distribution functions given by

$$p(k; \mu_1, \mu_2) \equiv P(Z = k) = \frac{1}{1 + \mu_1 + \mu_2} \begin{cases} \left(\frac{\mu_1}{1 + \mu_1}\right)^k, & k = 0, 1, 2, \dots \\ \left(\frac{\mu_2}{1 + \mu_2}\right)^{|k|}, & k = 0, -1, -2, \dots \end{cases} \quad (2)$$

and

$$P(Z \leq k) = \begin{cases} 1 - \frac{(1 + \mu_1 + \mu_2)^{-1} \mu_1^k}{(1 + \mu_1)^{k+1}}, & k = 0, 1, 2, \dots \\ \frac{(1 + \mu_2)^{k+1}}{(1 + \mu_1 + \mu_2) \mu_2^k}, & k = 0, -1, -2, \dots \end{cases},$$

respectively. The SDL distribution shares many of the properties of the skew (continuous) Laplace distribution such as infinitely divisibility, closure under geometric summation and a maximum entropy property. Moreover, a random variable following this distribution can be stochastically represented as a difference between two independent but not necessarily identically distributed geometric random variables. For more detail and other properties, see Kozubowski and Inusah (2006).

With the notations and definitions above, we are ready to introduce our SINARZ(1) process.

Definition 2 (*SINARZ(1) process*) Let $\{X_t\}_{t \in \mathbb{N}}$ and $\{Y_t\}_{t \in \mathbb{N}}$ be two independent NGI-NAR(1) processes with geometric marginals with mean $\mu_1 \geq 0$ and $\mu_2 \geq 0$ (respectively) and common parameter $\alpha > 0$ related to the counting series, as presented in Definition 1. More specifically, we define

$$X_t = \alpha * X_{t-1} + \epsilon_t, \quad t \in \mathbb{N}^*$$

and

$$Y_t = \alpha * Y_{t-1} + \nu_t, \quad t \in \mathbb{N}^*.$$

The sequences $\{\epsilon_t\}_{t \in \mathbb{N}^*}$ and $\{\nu_t\}_{t \in \mathbb{N}^*}$ are the innovations of the processes $\{X_t\}_{t \in \mathbb{N}}$ and $\{Y_t\}_{t \in \mathbb{N}}$ (respectively) and are defined as that one of Definition 1. Let $\{Z_t\}_{t \in \mathbb{N}^*}$ be a sequence of random variables following a common skew discrete Laplace distribution with parameters μ_1 and μ_2 , and define $\varepsilon_t = \epsilon_t - \nu_t$, for $t \in \mathbb{N}^*$. Then, we define our modified negative binomial thinning operator “ $\odot$ ” by

$$\alpha \odot Z_{t-1} \stackrel{d}{=} \alpha * X_{t-1} - \alpha * Y_{t-1},$$

for $t \in \mathbb{N}^*$. With this, we define completely our SINARZ(1) process $\{Z_t\}_{t \in \mathbb{N}}$ by

$$Z_t = \alpha \odot Z_{t-1} + \varepsilon_t,$$

for $t \in \mathbb{N}^*$.

Remark 1 From the results of Ristić et al. (2009), we have that the SINARZ(1) process is well-defined for $\alpha \leq \min\{\mu_1/(1 + \mu_1), \mu_2/(1 + \mu_2)\}$. With this, it is possible to find the distribution of ε_t , as will be discussed below. From now on, we consider the SINARZ(1) process with this restriction on α .

Remark 2 If $\mu_1 = \mu_2$, $\{Z_t\}_{t \in \mathbb{N}}$ is a symmetric INAR(1) process with symmetric discrete Laplace marginals (Inusah and Kozubowski 2006).

In Fig. 1 we present some simulated trajectories of the SINARZ(1) process for $(\mu_1, \mu_2) = (5, 5)$ (top plots) and $(\mu_1, \mu_2) = (10, 5)$ (bottom plots) with $\alpha = 0.4, 0.8$.

Marginal properties of our process can be obtained directly from the results given in Kozubowski and Inusah (2006). For example, the characteristic function of Z_t , denoted by $\phi(s) \equiv E(\exp(isZ_t))$ (with $i = \sqrt{-1}$), is given by

$$\phi(s) = \{[1 + \mu_1(1 - e^{is})][1 + \mu_2(1 - e^{-is})]\}^{-1}, \quad s \in \mathbb{R}.$$

The moments and absolute moments of Z_t are given by

$$E(Z_t^k) = \frac{(1 + \mu_1)(1 + \mu_2)}{1 + \mu_1 + \mu_2} \sum_{j=1}^k j! S(k, j) \left(\frac{\mu_1^j}{1 + \mu_2} + (-1)^k \frac{\mu_2^j}{1 + \mu_1} \right)$$

and

$$E(|Z_t|^k) = \frac{(1 + \mu_1)(1 + \mu_2)}{1 + \mu_1 + \mu_2} \sum_{j=1}^k j! S(k, j) \left(\frac{\mu_1^j}{1 + \mu_2} + \frac{\mu_2^j}{1 + \mu_1} \right), \quad (3)$$

respectively, where $S(k, j) = j!^{-1} \sum_{i=0}^{j-1} (-1)^i \binom{j}{i} (j-i)^k$ is the Stirling number of second kind. In particular, the expected value, variance and first absolute moment of Z_t are given by

$$\mu \equiv E(Z_t) = \mu_1 - \mu_2, \quad \sigma^2 \equiv \text{Var}(Z_t) = \mu_1(1 + \mu_1) + \mu_2(1 + \mu_2) \quad (4)$$

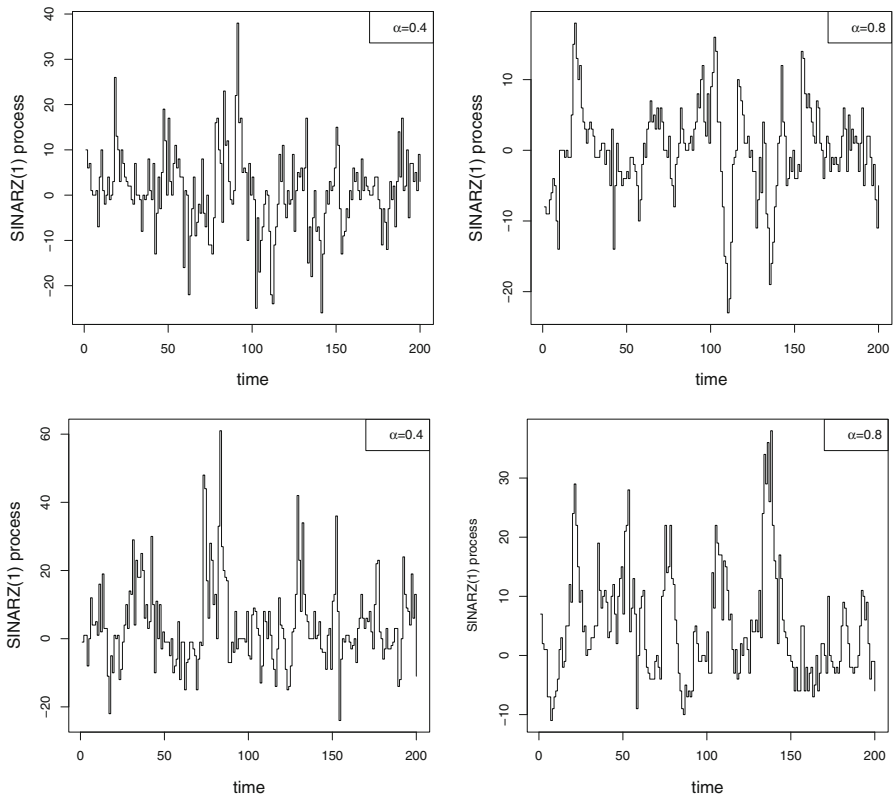


Fig. 1 Simulated trajectories of the SINARZ(1) process by setting $(\mu_1, \mu_2) = (5, 5)$ (top plots) and $(\mu_1, \mu_2) = (10, 5)$ (bottom plots) with $\alpha = 0.4, 0.8$

and

$$E(|Z_t|) = \frac{\mu_1(1 + \mu_1) + \mu_2(1 + \mu_2)}{1 + \mu_1 + \mu_2},$$

respectively.

With the restriction given in the Remark 1 and using the definition of ε_t and the result (1), we obtain that the probability function of ε_t can be expressed by

$$P(\varepsilon_t = k) = \beta_1 p(k; \mu_1, \mu_2) + \beta_2 p(k; \mu_1, \alpha) + \beta_3 p(k; \alpha, \mu_2) + (1 - \beta_1 - \beta_2 - \beta_3) p(k; \alpha, \alpha),$$

for $k \in \mathbb{Z}$, where $p(k; \cdot, \cdot)$ is defined in (2) and

$$\beta_1 = \left(1 - \frac{\alpha\mu_1}{\mu_1 - \alpha}\right) \left(1 - \frac{\alpha\mu_2}{\mu_2 - \alpha}\right), \quad \beta_2 = \left(1 - \frac{\alpha\mu_1}{\mu_1 - \alpha}\right) \frac{\alpha\mu_2}{\mu_2 - \alpha},$$

$$\beta_3 = \frac{\alpha\mu_1}{\mu_1 - \alpha} \left(1 - \frac{\alpha\mu_2}{\mu_2 - \alpha}\right).$$

That is, the random variable ε_t is distributed as a mixture of skew discrete Laplace random variables. Using this, it is straightforward to obtain that the characteristic function of ε_t , denoted by $\phi_\varepsilon(s) = E(\exp(is\varepsilon_t))$, is given by

$$\phi_\varepsilon(s) = \frac{[1 + \alpha(1 + \mu_1)(1 - e^{is})][1 + \alpha(1 + \mu_2)(1 - e^{-is})]}{[1 + \mu_1(1 - e^{is})][1 + \alpha(1 - e^{is})][1 + \mu_2(1 - e^{-is})][1 + \alpha(1 - e^{-is})]}, \quad (5)$$

for $s \in \mathbb{R}$. We have that the two first cumulants of ε_t are given by

$$\mu_\varepsilon \equiv E(\varepsilon_t) = (1 - \alpha)(\mu_1 - \mu_2), \quad (6)$$

$$\sigma_\varepsilon^2 \equiv \text{Var}(\varepsilon_t) = (1 + \alpha)\{\mu_1[(1 - \alpha)(1 + \mu_1) - \alpha] + \mu_2[(1 - \alpha)(1 + \mu_2) - \alpha]\}. \quad (7)$$

From our definition, we have that the SINARZ(1) process can be seen as a difference between two independent NGINAR(1) processes, that is, $Z_t \stackrel{d}{=} X_t - Y_t$, with X_t and Y_t as in Definition 1. From this, we can obtain some properties for our process from the properties of the NGINAR(1) process. The following proposition states some results that follow from this fact.

Proposition 1 *The following results are valid for the SINARZ(1) model:*

- (i) *It is Markovian, stationary and ergodic;*
- (ii) *The autocorrelation $\rho(k) = \text{Corr}(Z_t, Z_{t-k})$ is given by $\rho(k) = \alpha^k$, for $k \in \mathbb{N}$;*
- (iii) *The spectral density function $f(\omega) \equiv (2\pi)^{-1} \sum_{k \in \mathbb{Z}} \text{Cov}(Z_t, Z_{t-k}) e^{-ik\omega}$ reduces to*

$$f(\omega) = \frac{(1 - \alpha^2)}{2\pi} \frac{\mu_1(1 + \mu_1) + \mu_2(1 + \mu_2)}{1 + \alpha^2 - 2\alpha \cos(\omega)}, \quad \omega \in (-\pi, \pi].$$

Remark 3 From the proposition above, we see that our SINARZ(1) process has positive autocorrelation. It is possible to define an INAR(1) process having negative autocorrelation and with SDL marginals. For this, we follow the idea of [Freeland \(2010\)](#) and define

$$Z_t \stackrel{d}{=} \begin{cases} X_t - Y_t, & t = 0, 2, 4, \dots \\ Y_t - X_t, & t = 1, 3, 5, \dots \end{cases},$$

with $\varepsilon_t = \epsilon_t - \eta_t$ for $t = 0, 2, 4, \dots$ and $\varepsilon_t = \eta_t - \epsilon_t$ for $t = 1, 3, 5, \dots$. In this case, it can be shown that $\rho(k) = \text{Corr}(Z_t, Z_{t-k}) = (-\alpha)^k$, for $k \in \mathbb{N}$. Properties for this process follow in a similar fashion to that ones related to the SINARZ(1) model considered along this paper.

We now obtain the conditional characteristic function of $Z_t | Z_{t-1}$ and the joint characteristic function of (Z_t, Z_{t-1}) . The expressions of the associated probability functions are cumbersome and therefore omitted here.

Denote by N_r a random variable following a negative binomial distribution with parameters $r \geq 0$ and $\alpha > 0$ with probability function assuming the form

$$P(N_k = k) = \binom{k+r-1}{k} \left(\frac{1}{1+\alpha} \right)^r \left(\frac{\alpha}{1+\alpha} \right)^k,$$

for $k \in \mathbb{N}$. The associated characteristic function $\phi_{N_r}(s) \equiv E(\exp(isN_r))$ is given by

$$\phi_{N_r}(s) = [1 + \alpha(1 - e^{is})]^{-r},$$

for $s \in \mathbb{R}$. Let z be a non-negative integer. Using the stochastic representation of Z_t and the characteristic function above, we obtain that

$$\begin{aligned} E(\exp(isZ_t)|Z_{t-1} = z) &= \frac{\phi_\varepsilon(s)}{P(Z_{t-1} = z)} \sum_{y=0}^{\infty} P(X_{t-1} = z+y, Y_{t-1} = y) \\ &\quad \times \phi_{N_{z+y}}(s) \phi_{N_y}(-s) \\ &= \frac{(1+\mu_1+\mu_2)\phi_\varepsilon(s)[1+\alpha(1-e^{is})]^{1-z}[1+\alpha(1-e^{-is})]}{(1+\mu_1)(1+\mu_2)[1+\alpha(1-e^{is})][1+\alpha(1-e^{-is})]-\mu_1\mu_2}, \end{aligned}$$

for $s \in \mathbb{R}$, where $\phi_\varepsilon(\cdot)$ is the characteristic function of ε_t given in (5). For $z < 0$ integer, it can be shown in a similar way that the conditional characteristic function $E(\exp(isZ_t)|Z_{t-1} = z)$ can be expressed by

$$\begin{aligned} E(\exp(isZ_t)|Z_{t-1} = z) \\ = \frac{(1+\mu_1+\mu_2)\phi_\varepsilon(s)[1+\alpha(1-e^{is})][1+\alpha(1-e^{-is})]^{1+z}}{(1+\mu_1)(1+\mu_2)[1+\alpha(1-e^{is})][1+\alpha(1-e^{-is})]-\mu_1\mu_2}, \quad s \in \mathbb{R}. \end{aligned}$$

Hence, we obtain in particular that the mean and variance of Z_t given Z_{t-1} are given by

$$E(Z_t|Z_{t-1} = z) = \mu_\varepsilon + \alpha z$$

and

$$\text{Var}(Z_t|Z_{t-1} = z) = \sigma_\varepsilon^2 + \alpha(1+\alpha)|z| + \frac{2\alpha(1+\alpha)\mu_1\mu_2}{1+\mu_1+\mu_2},$$

respectively, for $z \in \mathbb{Z}$, with μ_ε and σ_ε^2 given in (6) and (6), respectively.

We now obtain an expression for the joint characteristic function of (Z_t, Z_{t-1}) , which we denote by $\phi(s, u) = E(\exp(isZ_t + iuZ_{t-1}))$, for $u, s \in \mathbb{R}$. We have that

$$\phi(s, u) = E\{\exp(iuZ_{t-1})E(\exp(isZ_t)|Z_{t-1})\}.$$

Hence, it can be shown that the double expectation in the right side of the above equation can be expressed by

$$E\{\exp(iuZ_{t-1})E(\exp(isZ_t)|Z_{t-1})\} = (1 + \mu_1 + \mu_2)\varphi(t)\{E(s^{*Z_t}1\{Z_t \geq 0\}) + E(s^{\dagger|Z_t|}1\{Z_t < 0\})\},$$

where

$$\varphi(s) = \frac{\phi_\varepsilon(s)[1 + \alpha(1 - e^{is})][1 + \alpha(1 - e^{-is})]}{(1 + \mu_1)(1 + \mu_2)[1 + \alpha(1 - e^{is})][1 + \alpha(1 - e^{-is})] - \mu_1\mu_2}, \quad (8)$$

$s^* = e^{iu}[1 + \alpha(1 - e^{is})]^{-1}$ and $s^\dagger = e^{iu}[1 + \alpha(1 - e^{-is})]^{-1}$. It can be checked that

$$E(s^{*Z_t}1\{Z_t \geq 0\}) = \frac{1 + \mu_1}{1 + \mu_1 + \mu_2} \frac{1 + \alpha(1 - e^{is})}{(1 + \mu_1)[1 + \alpha(1 - e^{is})] - e^{iu}\mu_1}$$

and

$$E(s^{\dagger|Z_t|}1\{Z_t < 0\}) = \frac{\mu_2}{1 + \mu_1 + \mu_2} \frac{e^{iu}}{(1 + \mu_2)[1 + \alpha(1 - e^{-is})] - e^{iu}\mu_2}.$$

With the results above we obtain that the joint characteristic function of (Z_t, Z_{t-1}) can be expressed by

$$\phi(s, u) = \varphi(s) \left\{ \frac{(1 + \mu_1)[1 + \alpha(1 - e^{is})]}{(1 + \mu_1)[1 + \alpha(1 - e^{is})] - e^{iu}\mu_1} + \frac{\mu_2 e^{iu}}{(1 + \mu_2)[1 + \alpha(1 - e^{-is})] - e^{iu}\mu_2} \right\}, \quad (9)$$

where $\varphi(\cdot)$ is defined in (8).

3 Higher-order moments and jumps

This section is devoted to find some additional statistical measures than those given in the previous section. We here obtain joint higher-order moments for our process Z_t and study the jump process, which is defined by $J_t \equiv Z_t - Z_{t-1}$, for $t \in \mathbb{N}^*$. We use the following notation for the higher-order moments:

$$\mu(s_1, \dots, s_{r-1}) \equiv E(Z_t Z_{t+s_1} \dots Z_{t+s_{r-1}}),$$

with $0 \leq s_1 \leq \dots \leq s_{r-1}$ and $r \geq 2$. In the following proposition, we present the second and third-order joint moments of Z_t (the first-order moment was presented in the previous section). This result can be obtained by using the stochastic representation of Z_t (that is, Z_t as a difference between two independent NGINAR(1) processes) and the results given in Theorem 1 from [Bakouch \(2010\)](#).

Proposition 2 *The second-order and third-order joint moments of the SINARZ(1) process are given by*

$$\mu(s) = \alpha^s [\mu_1(1 + \mu_1) + \mu_2(1 + \mu_2)] + (\mu_1 - \mu_2)^2,$$

$$\begin{aligned} \mu(0, s) = & 2(2 + \alpha^s) \mu_1 \mu_2 (\mu_2 - \mu_1) + \mu_1^2 (1 + 2\mu_1) - \mu_2^2 (1 + 2\mu_2) \\ & + \alpha^s [\mu_1(1 + 5\mu_1 + 4\mu_1^2) - \mu_2(1 + 5\mu_2 + 4\mu_2^2)], \end{aligned}$$

$$\begin{aligned} \mu(s, s) = & 2(2 + \alpha^s) \mu_1 \mu_2 (\mu_2 - \mu_1) + \mu_1^2 (1 + 2\mu_1) - \mu_2^2 (1 + 2\mu_2) \\ & + \frac{2\alpha^{s+1}}{1 - \alpha} [\mu_2(1 + \mu_2)^2 - \mu_1(1 + \mu_1)^2] + \frac{2\alpha^{2s}}{1 - \alpha} [\mu_1^2 (1 + \mu_1) - \mu_2^2 (1 + \mu_2)] \\ & + \frac{\alpha^{s+1}}{1 - \alpha} [\mu_1(1 + \mu_1)(1 - 2\mu_1) - \mu_2(1 + \mu_2)(1 - 2\mu_2)] \\ & + \frac{\alpha^s}{1 - \alpha} [\mu_1(1 + \mu_1)(1 + 2\mu_1) - \mu_2(1 + \mu_2)(1 + 2\mu_2)], \end{aligned}$$

$$\begin{aligned} \mu(s, u) = & -\mu_2 [3\mu_1^2 + \mu_1(1 + \mu_1)(\alpha^s + \alpha^u + \alpha^{u-s})] + \mu_1 [3\mu_2^2 + \mu_2(1 + \mu_2) \\ & \times (\alpha^s + \alpha^u + \alpha^{u-s})] \frac{\alpha^{u-s}}{1 - \alpha} \{2\alpha^{2s+1} [\mu_2(1 + \mu_2)^2 - \mu_1(1 + \mu_1)^2] \\ & + 2\alpha^{2s} [\mu_1^2 (1 + \mu_1) - \mu_2^2 (1 + \mu_2)] \\ & + \alpha^{s+1} [\mu_1^2 (1 + \mu_1^2)(1 - 2\mu_1) - \mu_2(1 + \mu_2)(1 - 2\mu_2)] \\ & + \alpha^s [\mu_1^2 (1 + \mu_1^2)(1 + 2\mu_1) - \mu_2(1 + \mu_2)(1 + 2\mu_2)] \\ & + \alpha^{u-s} [\mu_1^2 (1 + 2\mu_1) - \mu_2^2 (1 + 2\mu_2)] \\ & + (1 - \alpha^{u-s}) \{ \alpha^s [\mu_1^2 (1 + \mu_1) - \mu_2^2 (1 + \mu_2)] + \mu_1^3 - \mu_2^3 \}, \quad s < u. \end{aligned}$$

We now focus on the properties of the jump process $J_t = Z_t - Z_{t-1}$, $t \in \mathbb{N}^*$. Jump processes have been considered and studied in the literature due to the applications in checking the adequacy of fitted models. Further, they have been used to construct control charts to detect changes in the serial dependence structure, as proposed by Weiß (2009b). For instance, jumps in the Poisson and binomial count processes were investigated by Weiß (2008, 2009a) and Weiß (2009b), respectively.

Taking $u = -s$ in (9), we obtain that the characteristic function of J_t can be expressed by

$$\begin{aligned} E(\exp(isJ_t)) = \varphi(s) \left\{ \frac{(1 + \mu_1)[1 + \alpha(1 - e^{is})]}{(1 + \mu_1)[1 + \alpha(1 - e^{is})] - e^{-is}\mu_1} \right. \\ \left. + \frac{\mu_2 e^{-is}}{(1 + \mu_2)[1 + \alpha(1 - e^{-is})] - e^{-is}\mu_2} \right\}, \end{aligned}$$

where $\varphi(\cdot)$ is given in (8). We now present the first three moments of J_t . These results can be obtained directly or using the characteristic function above.

Proposition 3 *The first three moments of the jump process J_t are $E(J_t) = 0$, $E(J_t^2) = 2(1 - \alpha)[\mu_1(1 + \mu_1) + \mu_2(1 + \mu_2)]$ and $E(J_t^3) = 3\alpha[\mu_1(1 + 5\mu_1 + 4\mu_1^2) - \mu_2(1 + 5\mu_2 + 4\mu_2^2)] - 6\alpha^3(1 - \alpha)^{-1}[\mu_2(1 + \mu_2)^2 - \mu_1(1 + \mu_1)^2] - 3\alpha^2(1 - \alpha)^{-1}[\mu_1(1 + \mu_1) - \mu_2(1 + \mu_2)] - 3\alpha(1 - \alpha)^{-1}[\mu_1(1 + \mu_1)(1 + 2\mu_1) - \mu_2(1 + \mu_2)(1 + 2\mu_2)]$.*

We also obtain the autocorrelation function of J_t , which is denoted here by $\rho_J(\cdot)$. Using the autocorrelation function $\rho(\cdot)$ of Z_t , it can be shown that

$$\rho_J(k) = 2\rho(k) - \rho(k + 1) - \rho(k - 1) = -\alpha^{k-1}(1 - \alpha)^2,$$

for $k \in \mathbb{N}^*$. Note that the autocorrelation function of J_t is always negative.

4 Estimation and inference

We here propose estimators for the parameters of our process and find their asymptotic distributions. We do not here consider estimation by maximum likelihood since the likelihood function for our model is cumbersome to work with. Let n be the sample size of the time series Z_t . We start proposing an estimator for the parameter α based on the conditional least square method. In this case, the function to be minimized is given by

$$Q_n(\alpha, \mu) = \sum_{t=2}^n (Z_t - \alpha Z_{t-1} - (1 - \alpha)\mu)^2.$$

Note that this method does not provide estimators for μ_1 and μ_2 , but only for α and $\mu = \mu_1 - \mu_2$. For the symmetric case $\mu_1 = \mu_2$ ($\mu = 0$), we obtain that the estimator $\hat{\alpha}$ of α becomes

$$\hat{\alpha} = \frac{\sum_{t=2}^n Z_t Z_{t-1}}{\sum_{t=2}^n Z_{t-1}^2}.$$

Under the non-symmetric case $\mu_1 \neq \mu_2$ ($\mu \neq 0$), we obtain that the estimator $\hat{\alpha}$ of α is given by

$$\hat{\alpha} = \frac{(n - 1) \sum_{t=2}^n Z_t Z_{t-1} - \sum_{t=2}^n Z_t \sum_{t=2}^n Z_{t-1}}{(n - 1) \sum_{t=2}^n Z_{t-1}^2 - \left(\sum_{t=2}^n Z_{t-1} \right)^2}.$$

In the next proposition we establish the strong consistency and the asymptotic distribution of $\hat{\alpha}$, which are valid in both symmetric and non-symmetric cases.

Proposition 4 *The estimator $\hat{\alpha}$ is strongly consistent for α and satisfies the asymptotic normality*

$$\sqrt{n}(\hat{\alpha} - \alpha) \xrightarrow{d} N(0, v^2), \quad (10)$$

as $n \rightarrow \infty$, with

$$v^2 = \frac{1}{\sigma^2} \left(\sigma_\varepsilon^2 + \frac{2\alpha(1+\alpha)\mu_1\mu_2}{1+\mu_1+\mu_2} \right) + \frac{\alpha(\alpha+1)}{\sigma^4} [E(|Z|^3) - 2\mu E(\text{sgn}(Z)Z^2) + \mu^2 E(|Z|)],$$

where $\text{sgn}(Z) = 1$ if $Z \geq 0$ and $\text{sgn}(Z) = -1$ if $Z < 0$, $\sigma^4 \equiv (\sigma^2)^2$ and Z is a random variable following a SDL distribution with parameters μ_1 and μ_2 . The variances σ^2 and σ_ε^2 are given in (4) and (6), respectively.

Proof See Appendix. $\square$

Remark 4 The first and third moments of $|Z|$ involved in the asymptotic variance of $\hat{\alpha}$ can be obtained from (3). The expected value of $\text{sgn}(Z)Z^2$ is given by

$$E(\text{sgn}(Z)Z^2) = \frac{\mu_1(1+\mu_1)(1+2\mu_1) - \mu_2(1+\mu_2)(1+2\mu_2)}{1+\mu_1+\mu_2}.$$

We now move our attention to the estimation of μ_1 and μ_2 . As mentioned before, the conditional least square method does not provide estimators for these parameters, only for $\mu = \mu_1 - \mu_2$. To estimate μ_1 and μ_2 we here propose the method of moments based on the sample quantities of $E(Z)$ and $E(|Z|)$. Before to present explicitly the estimators, we introduce some notation that appears in Kozubowski and Inusah (2006) and that will be important for what follows.

Define two real functions $F_1(\cdot, \cdot)$ and $F_2(\cdot, \cdot)$ by

$$F_1(x, y) = \frac{2y + (x - y)(1 + \sqrt{1 + 4xy})}{(1 + \sqrt{1 + 4xy})(1 + x - y)}$$

and

$$F_2(x, y) = \frac{2y(1 + x - y)}{1 + 2y(x - y) + \sqrt{1 + 4xy}}.$$

With these definitions above, we immediately obtain from the proof of Proposition 5.2 of Kozubowski and Inusah (2006) that the estimators $\hat{\mu}_1$ and $\hat{\mu}_2$ of μ_1 and μ_2 (respectively) based on the method of moments (with the sample quantities of $E(Z)$ and $E(|Z|)$) are given by

$$\hat{\mu}_1 = \frac{F_1(\bar{Z}_n^+, \bar{Z}_n^-)}{1 - F_1(\bar{Z}_n^+, \bar{Z}_n^-)} \quad \text{and} \quad \hat{\mu}_2 = \frac{F_2(\bar{Z}_n^+, \bar{Z}_n^-)}{1 - F_2(\bar{Z}_n^+, \bar{Z}_n^-)}, \quad (11)$$

if $\bar{Z}_n \geq 0$, and

$$\hat{\mu}_1 = \frac{F_2(\bar{Z}_n^-, \bar{Z}_n^+)}{1 - F_2(\bar{Z}_n^-, \bar{Z}_n^+)} \quad \text{and} \quad \hat{\mu}_2 = \frac{F_1(\bar{Z}_n^-, \bar{Z}_n^+)}{1 - F_1(\bar{Z}_n^-, \bar{Z}_n^+)}, \quad (12)$$

if $\bar{Z}_n < 0$, where $\bar{Z}_n^+ = n^{-1} \sum_{i=1}^n Z_i^+$ and $\bar{Z}_n^- = n^{-1} \sum_{i=1}^n Z_i^-$. For $x \in \mathbb{R}$, we have defined $x^+ = \max(0, x)$ and $x^- = (-x)^+$.

We now present a proposition that deals with the asymptotic properties of the proposed estimators $\hat{\mu}_1$ and $\hat{\mu}_2$.

Proposition 5 *The estimators $\hat{\mu}_1$ and $\hat{\mu}_2$ given in (11) and (12) are strongly consistent for estimating μ_1 and μ_2 , respectively, and satisfy the asymptotic normality*

$$\sqrt{n}\{(\hat{\mu}_1, \hat{\mu}_2) - (\mu_1, \mu_2)\} \xrightarrow{d} N((0, 0), \Sigma),$$

as $n \rightarrow \infty$, where Σ is some finite covariance matrix.

Remark 5 We call attention that the asymptotic covariance matrix Σ is complicated to be obtained analytically and we do not considered this point here. We suggest to obtain Σ numerically through a Monte Carlo simulation, as made in the numerical results presented later.

Proof of Proposition 5 See Appendix. $\square$

From the proposition above, we obtain that the asymptotic distribution of $\hat{\mu} = \hat{\mu}_1 - \hat{\mu}_2$ is given by

$$\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N(0, \Sigma_{11} + \Sigma_{22} - 2\Sigma_{12}),$$

as $n \rightarrow \infty$, where Σ_{ij} is the element (i, j) of the matrix Σ . With this, we can construct a confidence interval for μ and test the null hypothesis $H_0 : \mu_1 = \mu_2$ against the alternative hypothesis $H_1 : \mu_1 \neq \mu_2$. So, we reject the null hypothesis if the value 0 does not belong to the confidence interval of μ .

We here present a small numerical experiment to evaluate the finite-sample performance of the estimators $\hat{\alpha}$, $\hat{\mu}_1$ and $\hat{\mu}_2$ of α , μ_1 and μ_2 , respectively. We set the sample sizes $n = 50, 100, 200, 400$ and the values of the parameters $\alpha = 0.1, 0.15, 0.2, 0.25, 0.3$ and $(\mu_1, \mu_2) = (3, 3), (6, 3)$. To evaluate the point estimation of the parameters we consider the empirical mean and standard error (SE). Another interest here is to assess the estimation of the second and third moments of the jump process J_t . The Monte Carlo simulation experiments are performed using the R programming language; see <http://www.r-project.org>. The number of Monte Carlo replications R considered here was $R = 5000$.

Tables 1 and 2 present the empirical mean and standard error of the estimates of the parameters of our model. From the results presented in these tables, we see that $\hat{\alpha}$, $\hat{\mu}_1$ and $\hat{\mu}_2$ are close to the true values of the parameters for the cases considered, which means that the estimators proposed in the previous section can be used effectively for estimation in the STINARZ(1) process. We also observe that the bias and the standard error of the estimates decrease as the sample size n increases, as expected. We also observe that the standard errors of $\hat{\mu}_1$ and $\hat{\mu}_2$ increase as α increases.

Table 3 gives us the empirical mean of the estimates of the second and third moments of J_t and their true values for some values of the parameters and $n = 50, 100, 200, 400$;

Table 1 Empirical means and standard errors (SE) of the estimates of the parameters for $(\mu_1, \mu_2) = (3, 3)$ and some values of α and n

n	α	$\hat{\alpha}$	$SE(\hat{\alpha})$	$\hat{\mu}_1$	$SE(\hat{\mu}_1)$	$\hat{\mu}_2$	$SE(\hat{\mu}_2)$
50	0.10	0.0730	0.1387	2.9464	0.6197	2.9795	0.6239
	0.15	0.1179	0.1407	2.9424	0.6413	2.9645	0.6443
	0.20	0.1659	0.1394	2.9542	0.6618	2.9397	0.6658
	0.25	0.2126	0.1395	2.9473	0.6922	2.9590	0.7009
	0.30	0.2599	0.1408	2.9269	0.7182	2.9464	0.7331
100	0.10	0.0873	0.0982	2.9729	0.4397	2.9835	0.4399
	0.15	0.1343	0.1010	2.9768	0.4705	2.9851	0.4638
	0.20	0.1842	0.1011	2.9810	0.4824	2.9718	0.4777
	0.25	0.2336	0.1026	2.9774	0.5014	2.9831	0.4995
	0.30	0.2800	0.0992	2.9675	0.5306	2.9684	0.5262
200	0.10	0.0931	0.0708	2.9925	0.3107	2.9910	0.3110
	0.15	0.1435	0.0706	2.9866	0.3233	2.9877	0.3229
	0.20	0.1927	0.0699	2.9839	0.3368	2.9893	0.3372
	0.25	0.2417	0.0703	2.9899	0.3524	2.9891	0.3529
	0.30	0.2902	0.0716	2.9758	0.3741	2.9901	0.3652
400	0.10	0.0971	0.0503	2.9965	0.2210	2.9960	0.2249
	0.15	0.1458	0.0510	2.9946	0.2284	2.9956	0.2337
	0.20	0.1978	0.0506	2.9862	0.2391	2.9943	0.2421
	0.25	0.2448	0.0486	2.9970	0.2475	2.9964	0.2522
	0.30	0.2946	0.0503	2.9922	0.2612	2.9914	0.2657

the true values of the moments are replicated for all values of n in order to facilitate the comparison between the estimated and true moments. We here denote $\mu_j^{(2)} = E(J_t^2)$ and $\mu_j^{(3)} = E(J_t^3)$ and their empirical means by $\hat{\mu}_j^{(2)}$ and $\hat{\mu}_j^{(3)}$, respectively. We see a good performance of the estimated second and third moments of J_t based on the estimators given in the previous section, since they are close to the true values of the moments for the case considered here, mainly as the sample size is large.

5 Application

We here show the usefulness of the SINARZ(1) process by applying it to a real time series of the Saudi stock market. We consider the daily difference between the close and open prices in 2012 of the Saudi Telecom. Since these differences belong to $\mathbb{Z}/10$, we work with the rescaled time series $(close\ price - open\ price) \times 10$, which we call *number of ticks*. The number of ticks belongs to $\mathbb{Z}$ and is our time series of interest here in this section. The data set can be obtained online at the site <http://www.tadawul.com.sa> and are listed in Table 4.

Table 5 displays some descriptive statistics of the number of ticks. We see that this time series contains negative integer values and therefore the usual count time series models can not be applied in this case. The time series data and their sample autocorrelation and partial autocorrelation functions are displayed in the Fig. 2.

Table 2 Empirical means and standard errors (SE) of the estimates of the parameters for $(\mu_1, \mu_2) = (6, 3)$ and some values of α and n .

n	α	$\hat{\alpha}$	$SE(\hat{\alpha})$	$\hat{\mu}_1$	$SE(\hat{\mu}_1)$	$\hat{\mu}_2$	$SE(\hat{\mu}_2)$
50	0.10	0.0752	0.1402	5.9405	1.0885	2.9464	0.7036
	0.15	0.1179	0.1378	5.9241	1.1167	2.9200	0.7284
	0.20	0.1662	0.1359	5.9332	1.1889	2.9305	0.7581
	0.25	0.2159	0.1364	5.9134	1.2093	2.9276	0.7767
	0.30	0.2620	0.1364	5.9033	1.2896	2.9168	0.8035
100	0.10	0.0896	0.0980	5.9780	0.7804	2.9839	0.5074
	0.15	0.1346	0.0995	5.9615	0.8100	2.9613	0.5183
	0.20	0.1830	0.0981	5.9675	0.8213	2.9809	0.5322
	0.25	0.2318	0.0964	5.9618	0.8650	2.9526	0.5551
	0.30	0.2818	0.0961	5.9790	0.9266	2.9614	0.5737
200	0.10	0.0933	0.0704	5.9878	0.5412	2.9827	0.3535
	0.15	0.1447	0.0697	5.9957	0.5744	2.9855	0.3656
	0.20	0.1914	0.0713	5.9893	0.5925	2.9792	0.3763
	0.25	0.2404	0.0693	5.9839	0.6121	2.9784	0.3912
	0.30	0.2903	0.0693	5.9791	0.6495	2.9836	0.4088
400	0.10	0.0957	0.0503	5.9924	0.3821	2.9882	0.2510
	0.15	0.1472	0.0502	5.9872	0.4043	2.9960	0.2517
	0.20	0.1962	0.0494	5.9930	0.4107	2.9878	0.2656
	0.25	0.2461	0.0501	5.9778	0.4335	2.9824	0.2782
	0.30	0.2953	0.0499	5.9844	0.4621	2.9880	0.2914

Figure 2 suggests that a first-order autoregressive model may be appropriate for fitting the time series considered here. Furthermore, the behavior of the series indicates that it can be generated by a mean stationary model.

We here also compare our SINARZ(1) model with the TINAR(1) introduced by Freeland (2010). In order to make a fair comparison, we here consider a skew version of the TINAR(1) process as discussed in Section 6 of Freeland (2010). With this, the TINAR(1) model considered here has marginals following a Skellam distribution with parameters $\lambda_1(1 - \beta)^{-1}$ and $\lambda_2(1 - \beta)^{-1}$ ($\lambda_1, \lambda_2 > 0, 0 < \beta < 1$), that is, the marginals are distributed as $Y_1 - Y_2$, where Y_1 and Y_2 are two independent Poisson random variables with mean $\lambda_1(1 - \beta)^{-1}$ and $\lambda_2(1 - \beta)^{-1}$, respectively, and β is the associated thinning parameter of this process. To estimate λ_1 and λ_2 , we use the sample quantities of $E(Z_t)$ and $\text{Var}(Z_t)$; for the skew version of the Freeland model we have that $E(Z_t) = (\lambda_1 - \lambda_2)(1 - \beta)^{-1}$ and $\text{Var}(Z_t) = (\lambda_1 + \lambda_2)(1 - \beta)^{-1}$. We estimate β through the conditional least square method, which yields the same estimator of that proposed here for our thinning parameter α .

In the Table 6 we present the estimates of the parameters and the following three goodness-of-fit statistics: RMS (root mean square error), MAE (mean absolute error) and MdAE (median absolute error), where the error is defined by $Z_t - E(Z_t|Z_{t-1})$, for $t = 2, \dots, n$, with Z_t and n being the time series and the sample size, respectively. In general it is expected that the better model to fit the data presents the smaller values for these quantities. For a good discussion of these statistics, we recommend the reader to the paper by Hyndman and Koehler (2006).

Table 3 True values and empirical means of the second and third moments of J_I for some values of the parameters and $n = 50, 100, 200, 400$

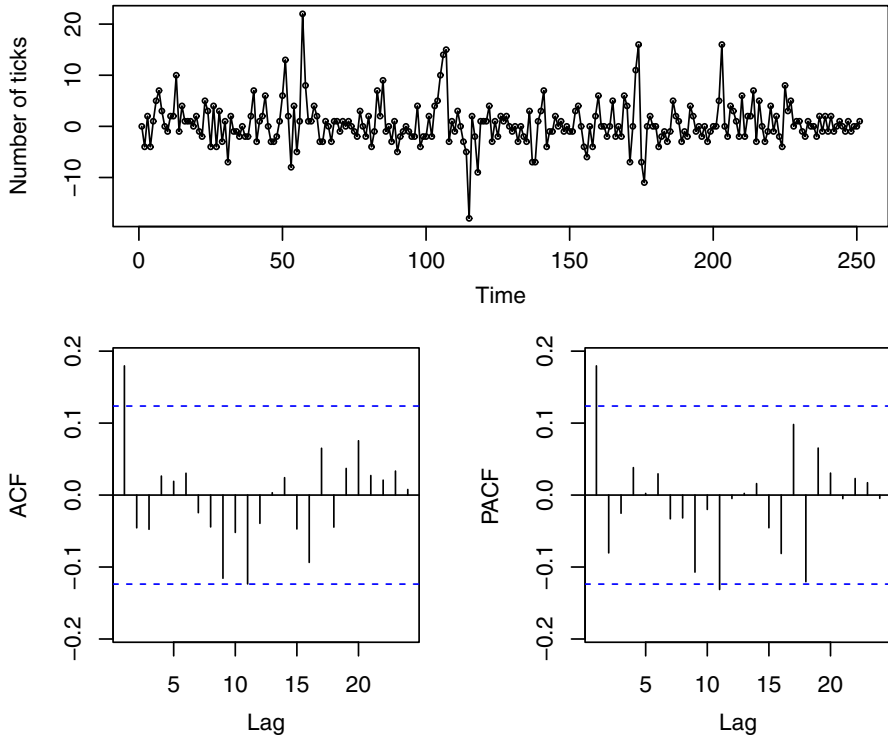
(μ_1, μ_2)		(3, 3)				(6, 3)			
n	α	$\mu_J^{(2)}$	$\hat{\mu}_J^{(2)}$	$\mu_J^{(3)}$	$\hat{\mu}_J^{(3)}$	$\mu_J^{(2)}$	$\hat{\mu}_J^{(2)}$	$\mu_J^{(3)}$	$\hat{\mu}_J^{(3)}$
50	0.10	43.2	44.8	0.0	0.396	97.2	100.3	114.8	72.3
	0.15	40.8	42.3	0.0	-0.261	91.8	95.2	161.2	116.3
	0.20	38.4	39.9	0.0	0.450	86.4	90.2	200.2	168.5
	0.25	36.0	37.8	0.0	-0.586	81.0	84.5	231.8	203.2
	0.30	33.6	33.6	0.0	0.638	75.6	79.4	255.9	237.5
100	0.10	43.2	44.0	0.0	0.075	97.2	98.9	114.8	92.4
	0.15	40.8	41.7	0.0	-0.116	91.8	93.5	161.2	139.6
	0.20	38.4	39.2	0.0	0.288	86.4	88.6	200.2	181.1
	0.25	36.0	36.8	0.0	-0.220	81.0	82.8	231.8	219.6
	0.30	33.6	33.9	0.0	-0.552	75.6	77.7	255.9	247.3
200	0.10	43.2	43.5	0.0	0.069	97.2	98.1	114.8	105.1
	0.15	40.8	41.1	0.0	0.044	91.8	92.7	161.2	154.7
	0.20	38.4	38.7	0.0	-0.284	86.4	87.5	200.2	192.1
	0.25	36.0	36.4	0.0	0.029	81.0	82.1	231.8	225.4
	0.30	33.6	33.9	0.0	0.026	75.6	76.1	255.9	249.9
400	0.10	43.2	43.4	0.0	0.048	97.2	97.9	114.8	108.2
	0.15	40.8	41.0	0.0	0.002	91.8	92.1	161.2	156.5
	0.20	38.4	38.4	0.0	-0.254	86.4	86.8	200.2	196.5
	0.25	36.0	36.3	0.0	0.002	81.0	81.1	231.8	227.5
	0.30	33.6	33.7	0.0	0.040	75.6	75.9	255.9	254.4

Table 4 The number of ticks of the Saudi Telecom in 2012

0	-4	2	-4	1	5	7	3	0	-1	2	2	10	-1	4	1	1	1
0	2	-1	-2	5	3	-4	4	-4	3	-3	1	-7	2	-1	-1	-2	0
-2	-2	2	7	-3	1	2	6	0	-3	-3	-2	1	6	13	2	-8	4
-5	1	22	8	1	1	4	2	-3	-3	1	0	-3	1	1	-1	1	0
1	0	-1	-2	3	0	-2	2	-4	-1	7	2	9	-1	0	-3	1	-5
-2	-1	0	-1	-2	-2	4	-4	-2	-2	2	-2	4	5	10	14	15	-3
1	-1	3	0	-3	-5	-18	2	-2	-9	1	1	1	4	-3	1	-2	2
1	2	0	-1	0	-3	0	-2	-3	3	-7	-7	1	3	7	-4	-1	-1
2	0	1	-1	0	-1	-1	3	4	0	-4	-6	0	-4	2	6	0	0
-2	0	5	-2	0	-2	6	4	-7	0	11	16	-7	-11	0	2	0	0
-4	-2	-1	-3	-1	5	2	1	-3	-1	-2	4	2	-1	0	-2	0	-3
-1	0	0	5	16	0	-2	4	3	1	-2	6	-2	2	2	7	-3	5
0	-3	-1	4	-1	2	-2	-4	8	3	5	0	1	1	-1	-2	1	0
0	-2	2	-1	2	-1	2	-1	0	1	0	-1	1	-1	0	0	1	

Table 5 Descriptive statistics for the number of ticks of the Saudi Telecom in 2012

Minimum	Median	Mean	Variance	$\hat{\rho}(1)$	Maximum
-18	0	0.5259	18.066	0.180	22

**Fig. 2** Plots of the time series, autocorrelation and partial autocorrelation functions for the number of ticks of the Saudi Telecom in 2012**Table 6** Estimates of the parameters and the goodness-of-fit statistics RMS, MAE and MdAE for the SINARZ(1) and TINAR(1) processes

Model	Estimates	RMS	MAE	MdAE
SINARZ(1)	$\hat{\alpha} = 0.1795$			
	$\hat{\mu}_1 = 2.6386$	4.181284	2.828201	2.021247
	$\hat{\mu}_2 = 2.1127$			
TINAR(1)	$\hat{\beta} = 0.1795$			
	$\hat{\lambda}_1 = 7.6272$	4.181284	2.828202	2.021247
	$\hat{\lambda}_2 = 7.1957$			

From the Table 6 we see that the SINARZ(1) and TINAR(1) processes provide similar fits to the data based on the goodness-of-fit statistics RMS, MAE and MdAE.

The standard errors for the estimates of the parameters α , μ_1 and μ_2 are respectively 0.0645, 0.2629 and 0.2252. The estimated covariance between $\hat{\mu}_1$ and $\hat{\mu}_2$ is 0.0125; the standard errors and covariance of $\hat{\mu}_1$ and $\hat{\mu}_2$ are obtained via Monte Carlo

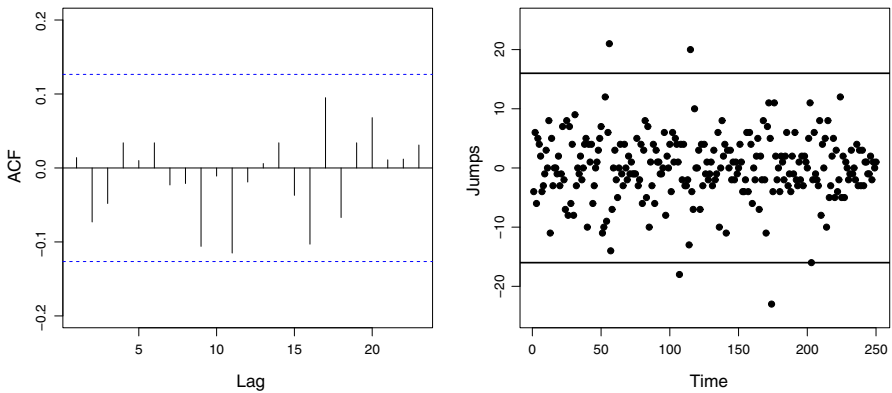


Fig. 3 Sample autocorrelation function of the residuals and jumps against time of our fitted SINARZ(1) model

simulation. We also obtain confidence intervals with a significance level at 5% for the parameters α , μ_1 and μ_2 , which are given by (0.0531; 0.3059), (2.6061; 2.6711) and (2.0848; 2.1406), respectively. In order to test the null hypothesis $H_0 : \mu_1 = \mu_2$ against the alternative hypothesis $H_1 : \mu_1 \neq \mu_2$, we construct a confidence interval (with a significance level at 5%) for μ as proposed in the Sect. 4. The confidence interval for μ is (0.4878; 0.5640) and since it does not contain the value 0 we reject the null hypothesis in favor of the alternative hypothesis that the data were generated by a SINARZ(1) model with $\mu_1 \neq \mu_2$.

In the Fig. 3 we present plots of the sample autocorrelation function of the residuals and the jumps against time (Shewhart control chart) with $\pm 3\sigma_J$ limits chosen as the benchmark chart as proposed by Weiß (2009b); here we define $\sigma_J = \sqrt{\text{Var}(J_t)}$, where the variance of J_t is given in Proposition 3. These plots indicate that the residuals are not correlated and that there is not a particular point causing a huge impact in the model. Therefore, our SINARZ(1) model seems to be well fitted to the number of ticks.

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Appendix

Proof of Proposition 4 It is straightforward to check that the conditions of Theorem 3.1 and Theorem 3.2 of Tjøstheim (1986) are satisfied in our case. Therefore, from these theorems we obtain respectively that $\hat{\alpha}$ is strongly consistent and that it satisfies the asymptotic normality given in (10).

The remaining point that needs to be shown is the expression of the asymptotic variance of $\sqrt{n}(\hat{\alpha} - \alpha)$. Following the notation of the paper by Tjøstheim (1986), we

have that $\tilde{Z}_{t|t-1} \equiv E(Z_t|Z_{t-1}) = \alpha Z_{t-1} + (1 - \alpha)\mu$. Hence, we get

$$\begin{aligned} \frac{\partial \tilde{Z}_{t|t-1}}{\partial \alpha} &= Z_{t-1} - \mu, \quad U \equiv E \left[\left(\frac{\partial \tilde{Z}_{t|t-1}}{\partial \alpha} \right)^2 \right] = \sigma^2, \\ f_{t|t-1} &\equiv E[(Z_t - \tilde{Z}_{t|t-1})^2 | Z_{t-1}] = \text{Var}(Z_t | Z_{t-1}) = \sigma_\varepsilon^2 + \alpha(1 + \alpha)|Z_{t-1}| \\ &\quad + \frac{2\alpha(1 + \alpha)\mu_1\mu_2}{1 + \mu_1 + \mu_2}, \\ &\equiv E \left[\left(\frac{\partial \tilde{Z}_{t|t-1}}{\partial \alpha} \right)^2 f_{t|t-1} \right] \\ &= \sigma^2 \left(\sigma_\varepsilon^2 + \frac{2\alpha(1 + \alpha)\mu_1\mu_2}{1 + \mu_1 + \mu_2} \right) \\ &\quad + \alpha(\alpha + 1)[E(|Z|^3) - 2\mu E(\text{sgn}(Z)Z^2) + \mu^2 E(|Z|)]. \end{aligned}$$

From the Theorem 3.2 of [Tjøstheim \(1986\)](#), we obtain that the asymptotic variance of $\sqrt{n}(\hat{\alpha} - \alpha)$ is given by R/U^2 . Using the expressions of R and U above, we obtain the desired result. $\square$

Proof of Proposition 5 The proof of this proposition follows the same steps than the proof of the Theorem 5.2 from [Kozubowski and Inusah \(2006\)](#) with some adaptations regarding the convergence aspects. We now discuss this in some detail.

We have that the stationarity and ergodicity properties of Z_t imply in the same properties for Z_t^+ and Z_t^- ; remember $x^+ = \max(0, x)$ and $x^- = (-x)^+$. Hence, we obtain that the consistency of the estimators $\hat{\mu}_1$ and $\hat{\mu}_2$ are obtained by using the same steps of the proof of the Theorem 5.2 from [Kozubowski and Inusah \(2006\)](#) and using the Law of Large Numbers for stationary and ergodic processes (for instance, see Proposition 7.1 from [Hamilton 1994](#)) instead of the classical Law of Large Numbers. The asymptotic normality of the estimators follow in the same way that in [Kozubowski and Inusah \(2006\)](#) by using the Central Limit Theorem for stationary processes (see Chapter 7 from [Anderson \(1971\)](#)) instead of classical Central Limit Theorem. $\square$

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